Worksheet answers

1. Which environment has the coldest average surface temperature?

   Titan (-180 °C)

2. Which environment has the greatest atmospheric pressure?

   Venus (about 90 bar)

3. Which solar system body has an average surface temperature that is most similar to Earth's?

   Mars

4. Describe differences between the Earth's atmosphere now and four billion years ago.

   The Earth's atmosphere today is composed of 77% nitrogen, 21% oxygen, and 2% of other gases. The atmospheric composition of Earth 4 billion years ago shows higher amounts of carbon dioxide (45%) and less nitrogen (10%).

5. Titan contains compounds that scientists believe are similar to the building blocks of early life on Earth. What differences exist between Earth and Titan that prevent us from assuming there is life on Titan?

   Life on Earth exists in conditions that include an average surface temperature of 15 °C and an atmosphere which includes oxygen.

   Titan's average surface temperature is -180 °C, which is too cold for most life forms on Earth. Titan also appears to have no oxygen, which is required by many life forms on Earth.

   One of the reasons for Titan's very cold temperature is that it is much further away from the Sun than the Earth (it is one of Saturn's moons). This means much less sunlight reaches the surface of Titan. This might affect the survival of life forms such as plants.

6. If found, what type of life is most likely to exist on Mars?

   Life on Mars would have to be well adapted to cold temperatures and high amounts of carbon dioxide. Perhaps a primitive form of plant life might exist on Mars, as conditions seem to be similar to those of Earth 4 billion years ago.

7. What challenges would humans be faced with if they decided to colonise:

   (a) Mars
   Temperature: it is very cold (-65 °C) and humans would need specialised living conditions, maybe similar to those used in Antarctica. A lack of breathable atmosphere would require some means of producing oxygen, and extracting water from ground-based frozen sources might be a challenge.

   (b) Titan
   Titan is extremely cold (-180 °C). This would require specialised living conditions because we don't find conditions this extreme on Earth. The atmosphere is not breathable by humans and might contain poisonous gases. Water might be difficult to find, but perhaps not impossible (it could be frozen). Sunlight would be poor, and it might be difficult to cultivate plants for food as there is no carbon dioxide in the atmosphere.

   (c) Mercury
   Mercury's average temperature is 180 °C, hotter than anywhere on Earth. Shielding humans from this extreme temperature and the solar radiation this close to the Sun would be a challenge. There is almost no atmospheric pressure on Mercury, so finding breathable air or water for drinking would probably be impossible. The temperature on Mercury means that there is probably no water to be found on the surface.

spice secondary teachers' enrichment program (science)

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